



SOLITUDE 16 MW SOLAR PV



Document Prepared by Earthood Services Private Limited

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Prepared By	Earthood Services Private Limited
Contact	Address: 1203-1205, 12th Floor, Tower B, Emaar Digital Greens, Sector 61, Gurugram, Haryana, India 122011 Contact no.: +91 124 420 4599 Email: info@earthood.in

	Website: www.earthood.in
Approved By	Dr. Kaviraj Singh Managing Director
Work Carried Out By	Team Leader: Anjali Chaudhary Validator/Verifier: Anjali Chaudhary Verifier (trainee): Kerwyn Dsouza Technical Expert (TA 1.2): Shifali Guleria Technical Reviewer (TA 1.2): Shreya Garg

Summary:

The project activity involves the generation of electricity through a greenfield solar photovoltaic plant with a capacity of 16.328 MWp in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. The solar plant consists of 59,376 photovoltaic modules that supply the generated electricity to the Central Electricity Board (CEB) grid.

Earthood Services Private Limited (Earthood) is contracted to conduct the third verification of the VCS registered project activity Solitude 16 MW Solar PV Project by the project proponent Voltas Yellow. The assessment is performed in line with the VCS requirements for project activities. The review of the MR, supporting documentation and subsequent follow-up actions (telephonic interviews with stakeholders) have provided Earthood with sufficient evidence to determine the fulfilment of stated criteria including but not limited to the VCS Standard /5/, VCS Validation and Verification Manual and other relevant rules and requirements established by Verra.

During the current monitoring period, from 01/01/2022 to 31/12/2022, the plant has successfully delivered 23,764 MWh of electricity to the national grid and the actual emission reductions were calculated as 23,557 tCO₂e.

A risk-based approach has been followed to perform this verification assessment and no uncertainties are associated with the verification. A total of 1 CLs, 2 CARs has been raised during the verification process of the project activity and successfully closed as detailed in appendix IV of this report.

Earthood Services Private Limited can certify that the emission reductions achieved in the for the monitoring period 01/01/2022 to 31/12/2022 by VCS project activity “Solitude 16MW Solar PV Project” (VCS ID:2308) amount to 23,557 tCO₂e. Therefore, this is being submitted for request of issuance, as per VCS procedures.

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1 INTRODUCTION

1.1 Objective

Voltas Yellow Ltd (PP) has contracted Earthood Services Private Limited (VVB) to conduct the verification of the project activity “Solitude 16MW Solar PV Project” (VCS ID:2308) located in Mauritius against the requirements of VCS Program.

The objective of the verification assessment is to determine that the

- The project complies with all the applicable VCS requirements of the current version of the VCS Standard v4.4 and program guide v4.3/05/.
- Project conformity with the registered monitoring plan/01/ and applied methodology, ACM0002: Grid-connected electricity generation from renewable sources — Version 20/06/
- The data/information reported is accurate, complete, consistent, transparent, and free of material error or omission.

In order to fulfil the above objectives, the VVB assigned the task of assessment to validation team that is collectively qualified as per Annex A.2.3.3 of ISO 14064-3 and other internal requirements.

The verification is an independent assessment of the project by a verification body that determines whether the project complies with the VCS rules and requirements.

1.2 Scope and Criteria

The verification scope of this assignment is defined as an independent and objective review of the VCS Monitoring report and the supporting annexures, which is reviewed by the assessment team against the relevant criteria including but not limited to monitoring methodology, applicable tools, VCS standard and guidance. The assessment team has applied a risk-based approach as per recommendations prescribed in the latest version of VCS Validation and Verification Manual.

The scope of this assignment is to:

- Conduct verification assessment in accordance with Earthood’s own QMS that is based on the VCS VVM along with the guidance provided by the VCS board to determine if the project meets all applicable VCS requirements, including those specified in the VCS standard, relevant methodology, tools, and guidelines and processing the same with VCS Program Guide.
- Assess the accuracy, conservativeness, relevance, completeness, consistency, and transparency of the information provided by the project proponents.
- Determine whether information provided by the project proponents is reliable and credible.
- Present information in the form of verification report in a factual, neutral, coherent manner and referencing the sources and assumptions, and other forms of verification means employed.
- Report the findings and conclusions in an objective manner.
- Safeguard the confidentiality of all information obtained during verification; and
- Adhere to the principles of independence, ethical conduct, fair presentation, and due professional care in assessment process.

1.3 Level of Assurance

- ☐ Limited level of assurance
- ☒ Reasonable level of assurance

Earthood's verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the verification by obtaining evidence and other information and explanations that Earthood considers necessary to give reasonable level of assurance that reported estimated GHG emission reductions are fairly stated.

In our opinion the estimated and actual GHG emissions reductions were calculated correctly on the basis of the registered monitoring plan/01/ and monitoring methodology/06/.

The assessment result has a reasonable level of assurance in verification that GHG assertions are free of material errors, omissions, and misrepresentations. The documents and evidence reviewed are included under Appendix II of this report.

1.4 Summary Description of the Project

Solitude 16 MW solar PV project consists of the construction and operation of a greenfield solar photovoltaic power plant in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses. The total installed capacity of solar PV plant amounts to 16.328 MW_{DC} which will capture solar energy and evacuate the produce electricity to Mauritius national grid.

The verification team based on the assessment of the documents, physical inspection during the site visit and interview with the PP and the plant personnel was able to confirm that the project activity is fully functional and implemented as described in the registered VCS PD/01/.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification of the project consists of following steps:

Desk review-

- Review of data and information provided in the MR/03/ and corresponding annexures.
- Cross check the information with the sources without limitations to the information provided by the project proponent.

Follow up actions-

- On site audit to inspect the project design, including, but not limited to, implementation status and monitoring mechanism.

- Interview with relevant personnel and local stakeholders responsible for information given in the VCS MR.

Reporting of findings-

- Resolution of findings
- Draft verification report

Independent technical review-

The project documentation is reviewed by internal technical reviewer in order to independently confirm whether the applicable GHG program requirements were objectively met or not, in addition to whether internal procedures were followed while arriving at the validation and verification opinion. The technical reviewer may accept or reject the verification opinion prepared by the assessment team and gives the reasons. The resolved findings may be opened at this stage or new findings may be identified that are required to be addressed by assessment team and/or project proponents, as appropriate. The technical reviewer is the decision maker on behalf of the Earthood Services Private Limited. A positive opinion is issued if all the findings have been satisfactorily resolved and in all other cases a negative opinion is issued unless the contract is terminated by either party before reaching the final opinion. Earthood keeps all documents and records in a secure and retrievable manner for at least two years after the end of the project crediting period.

2.2 Document Review

The verification is performed primarily as a document review of the documents submitted at various stages of assessments. The review is performed by assessment team using dedicated protocols. The assessment team cross checks the information provided in the documents, Project Description/01/, ER sheet/04/ & Monitoring Report/03/ and information from sources other than those used, if available, and also conducts independent background investigations. Earthood conducted a desk review as under:

- A. A review of the data and information presented to verify their completeness.
- B. A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures.
- C. An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

During the current verification period, an onsite visit was conducted to the plant site on 17/05/2023 in order to verify the implementation of the project activity. The on-site interviews with the PP representatives were conducted to discuss aspects of project implementation such as monitoring plan, the operations at

the project site (outages, if any), and other details like calibration of the monitoring equipment used during the current monitoring period, etc.

The required documents and details for the verification of project activity implementation were collected from project developer, through follow up emails, Microsoft teams video calls and interviews.

The interviews were conducted in English and Hindi; therefore no Local Expert requirement was identified by the assessment team for the onsite audit.

The details of the key personnel interviewed, and the main topics of the interviews are as follows-

S.no	Name	Affiliation	Topics discussed	Assessment team
1.	Josian Arlanda	Site Supervisor, GreenYellow Indian Ocean	- Project Implementation status -Operation and Maintenance of Facility, Monitoring practices, Data collection, data storage, QA/QC	Anjali Chaudhary, Shifali Guleria, Kerwyn Dsouza
2.	Marion Michot,	CSR/HSE Coordinator, GreenYellow Indian Ocean	- Roles &Responsibilities -Calibration requirements	
3.	Ramchand Coonjul	Technical Director- PV O&M, GreenYellow Indian Ocean	-GHG calculations -Ongoing grievance mechanism -Awareness programmes undertaken by PP	
4.	Gopika Callychurn,	PV Operations Analyst, GreenYellow Indian Ocean	Non-Involvement in Emissions Trading Programs and Other Binding Limits or Other Forms of Environmental Credit.	
5.	Pradeep Palaa	Local stakeholders	Ongoing grievance mechanism and feedback	
6.	Vijay Chaudhuriee			

2.4 Site Visits

During this verification, ESPL has conducted an onsite visit on 17/05/2023 to achieve the reasonable level of assurance that is in conformance with VCS rules. The details of the interviews conducted, and on-site activities undertaken have been described under section 2.3 above.

2.5 Resolution of Findings

The objective of this step is to identify, discuss and conclude on the issues related to the monitoring, implementation and operations of the registered project activity that could impair the capacity of the

registered project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions. This is done based on the desk review and onsite assessment. The verification team prepares and/or updates a verification protocol (internal document) that records the conformities and nonconformities, which may be of following types:

CAR (Corrective Action Request) is raised if one of the following occurs: Non-compliance with the monitoring plan, the methodology or the standardized baseline are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient.

Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions, change to the key sustainable development indicators. Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

Clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable requirements have been met. All CARs and CLs raised by the Earthhood during verification shall be resolved prior to submitting a request for issuance.

FAR (Forward Action Request) is raised during verification if the monitoring and reporting require attention and/or adjustment for the next verification period.

The total of 01 CL, 02 CARs was raised in the current verification and 01 FAR raised during the previous verification was successfully address during the current verification assessment. No FAR was raised during this monitoring period. All the findings that are raised and communicated to project participant during the verification are included under Appendix IV. The section also includes the response, if provided, by the project participants and an assessment by the verification team if it was closed out or otherwise.

2.5.1 Forward Action Requests

No FAR is raised.

2.6 Eligibility for Validation Activities

No validation activity is undertaken; thus, this is not applicable.

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

Project is registered under VCS only (VCS ID-2308)/07/. It is confirmed from the declaration/8/ provided by the project owner of “Solitude 16MW Solar PV Project” that the net GHG emission reductions generated by the project activity will not be used for compliance with any other emissions trading program or to meet binding limits on GHG emissions for the same monitoring period.

The project activity is not a part of any other GHG Program. The same has been verified from the CDM and GS websites.

3.2 Methodology Deviations

No methodology deviation has been applied for this monitoring period. Therefore, this section is not applicable to the project

3.3 Project Description Deviations

No deviation has been applied for this monitoring period compared to the project description. Therefore, this section is not applicable.

3.4 Grouped Project

This is not a group project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

The project activity consists of the construction and operation of a greenfield 16.328 MW_{DC} solar photovoltaic power plant in Solitude near the village of Triolet in the north of Mauritius, in the district of Pamplemousses.

The project has been operational since 19/12/2018. Under the third monitoring period from 01/01/2022 – 31/12/2022, the project has delivered 23,764 MWh of electricity to the national grid as verified from the invoices raised to CEB /28/, thereby offsetting an equivalent amount of electricity generated from fossil fuel-based plants, which has resulted in corresponding emission reductions of 23,557 tCO_{2e}.

It has also been verified as per the guidelines in ACM0002 Version 20 that the calculations for the GHG emission reductions is done in accordance with the aforesaid methodology/06/. The project activity is not a part of any other GHG allowance trading program, and the concerned monitoring period was not found overlapping with any of the VCS or CDM verifications so far. The total emission reductions applied for the current monitoring period under VCS are 23,557 tCO_{2e} and are not accounted in any other program. Therefore, the assessment team concluded that all physical features of the project activity in the registered PD are in place and the project participant has operated the project activity as per the registered PD /01/. Assessment team concludes the following:

1. There are no material discrepancies between project implementation and the project description provided in the registered PD/01/.
2. The monitoring plan is implemented completely and monitoring system (i.e., process and schedule for obtaining, recording, compiling, and analysing the monitored data and parameters) is appropriate.
3. There is no material discrepancies between the actual monitoring system, and the monitoring plan set out in the project description and the applied methodology/06/.

4. The project activity is registered confirmed been registered with CDM Ref No:10483/22/. It has been confirmed by the assessment team that PP is not claiming emission reductions for the same monitoring period in both the programmes. PP has confirmed and provided a no double counting declaration for the same/8/.

5. The project has not received or sought any other form of environmental credit or has become eligible to do so since validation or previous verification.

6. The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading/8/.

7. The project activity is complying with indicators for sustainable development as discussed in detail under section 1.11 of MR.

23,764 MWh of electricity to the national grid as verified from the invoices raised /28/ which has resulted in a reduction of 23,557 tCO₂e GHG emissions. The assessment team conducted a site visit to verify the implementation of the monitoring plan as per the registered PD/1/. It was confirmed through the interview of the PP and monitoring personnel that the quantity of net electricity generation supplied by the project plant to the grid is calculated as difference between:

(a) the quantity of electricity supplied by the project plant/unit to the grid.

(b) the quantity of electricity the project plant/unit from the grid, both reliably monitored and invoiced based on calibrated EDM1 electricity meters.

The SDG contributions are listed under section 1.11 of the MR, and the contributions were confirmed with the PP during the remote audit. The main sustainable development contributions have been listed under Energy security and supply, Health, Employment opportunities and Technology transfer.

4.2 Safeguards

4.2.1 No Net Harm

No negative environmental and social impacts have been identified because of the project development and implementation, according to the EIA Impact Assessment report/19,23/. The third party bi-annual Environmental Monitoring report for FY2022/20,21/ has been reviewed by the VVB to confirm that the operation of the project activity does not have any negative impact.

The project activity contributes to the following SDG as mentioned under section 1.11 of the Monitoring report.

- SDG 7: Indicator 7.2, Renewable energy share in the total final energy consumption. During the current MP the project activity delivered 23,764 MWh of solar PV generated renewable energy to the national grid of the Mauritius/28/.
- SDG 7: Indicator 7.b, Increase the accessibility of clean electricity to households. The project estimates to bring clean and sustainable electricity to 10,000 Mauritian households.
- SDG 8: Indicator 8.3 Offer employment opportunities. During the operation phase of the project activity, 10-15 workers were employed during the current Monitoring Period as verified from the staff log during the onsite assessment.
- SDG 4: Indicator 4.7, Provide awareness to younger generation about climate change and energy transition. In December 2022, 100 students were welcomed to the PV plant after having participated in a Climate Fresk Workshop to understand the stakes of global warming and how solar energy could tackle it. During the site visit, through interview with the CSR Coordinator, it was confirmed that that social awareness activities were conducted during the current Monitoring Period.
- SDG 1: Indicator 1.4 Help population in need. A solidarity collection among GreenYellow employees was organized to donate clothes, toys, and books to the association and to the children helped by it. In total, about 200 kg of donations were collected. It was confirmed by the assessment team through the interviews with the CSR Coordinator during the site visit that social commitment activities were conducted during the current Monitoring Period.
- SDG 12: Indicator 2.4, The assessment team observed during the site visit that farming activities were undertaken within the boundary of the solar PV plant. Thereby preventing runoff and social erosion and allowing maximum land utilisation.
- SDG 13: Indicator 13.0: Tonnes of greenhouse gas emissions avoided or removed; the project activity led to generation of renewable energy thereby leading to a total of 23,557 tCO_{2e} emission being reduced as verified from the calculation in the ER sheet/04/.

4.2.2 Local Stakeholder Consultation

As part of the public consultation process during EIA, a Local Stakeholder Consultation (LSC) meeting was held at the site office of Voltas Yellow Ltd on 06/07/2018, with the objective of gathering comments and concerns of stakeholders.

During the physical site visit, it was confirmed by PP through interviews and also cross-checked by assessment team that a grievance logbook is maintained at the plant site and no grievance has been registered during the current monitoring period/32/. The local stakeholders were interviewed as mentioned in section 2.3 above and it was confirmed that there is no negative sentiment in the local community with respect to plant operations.

4.3 AFOLU-Specific Safeguards

Not applicable since this project is a non-AFOLU project.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

The emission reductions are based on the introduction of renewable energy (solar power plants) by the project activity.

The emission reduction calculation is accurate from the point of data collection, transfer and the formulas applied are in accordance with the applied methodology/06/ and registered PD/01/.

The default values applied for calculations were consistent to the registered PD/01/.

The data transfer from the monthly generation sheets to final emission reduction calculation sheet/04/ has been transparently described and followed. The final values are reproducible, and all the calculation are linked and clearly presented in the emission reduction calculation sheet/04/.

The assessment team is able to conclude that the monitoring for the concerned period has been done in accordance with the procedures laid in the registered documents and the resulting emission reductions are measurable and conservative.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

The emission reductions calculations for this verification were verified as per the monitoring plan against all the necessary evidence as listed under Appendix II of the verification report, that were found to be appropriate and reliable sources of information. The information was also cross checked through the received documents and interviews conducted and observations made during the remote site visit conducted by the assessment team.

The assessment team confirms that appropriate methods and formulae for calculating baseline emissions have been followed. The assumptions, emission factors and default values that were applied in the calculations are justified. The actual emission reduction achieved during the current monitoring period is lesser than the estimated amount of emission reductions at the time of validation, therefore no further assessment was required. All the data were made available and have been monitored as per the required monitoring frequency. The means of verification for the values of parameters, used for baseline emission calculation, is described above in previous sections.

Parameters available at the time of Validation:

Parameter	Value	VVB Assessment
EF _{grid,y}	0.9915 tCO ₂ /MWh	CO₂ emission factor of the grid is calculated in accordance with “Tool to calculate the emission factor for an electricity system”/11/. The value shall be constant for the entire first crediting period. The value was found to be consistent with the PD/01/, across the MR/03/ and the ER sheet/04/.
EF _{grid,OM,y}	1.0282 tCO ₂ /MWh	Operating Margin CO₂ emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system". Source of data is IGES List of Grid Emission Factors. The value was found to be consistent with the registered CDM PDD/02/, registered VCS PD/01/ the ER sheet/04/ and is fixed ex-ante for the project activity.
EF _{grid,BM,y}	0.8814 tCO ₂ /MWh	Build Margin CO₂ emission factor for grid connected power generation in year y calculated using the CDM "Methodological Tool to calculate the emission factor for an electricity system". Source of data IGES List of Grid Emission Factors. The value was found to be consistent with the registered CDM PDD/02/, across registered VCS PD/01/, MR/03/ and the ER sheet/04/.
The percentage share of total installed capacity of solar PV	2.6%	The percentage share of total installed capacity of the solar PV in the total installed grid connected power generation capacity in the host country. Source of data- http://publicutilities.govmu.org/ The value is consistent with registered CDM PDD/02/ and registered VCS PD/01/ and fixed ex-ante for the project activity.

The total installed capacity of solar PV	21.2 MW	The total installed capacity of grid-connected solar PV in the host country. Source of data http://publicutilities.govmu.org/ The value is consistent with CDM PDD/02/ and registered VCS PD/01/ across the MR/03/.
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Parameters monitored during the Verification:

Parameter	EGfacility,y Quantity of net electricity generation supplied by the project plant/unit to the grid in year y									
Means of verification	Criteria/ Requirements	Assessment/observations								
	Measuring/Reading/Recording Frequency	The parameter is monitored continuously and recorded monthly for the estimation of GHG Emission Reductions. The parameter is being monitored continuously through the main EDM1 meter with Accuracy class 0.2S installed at CEB JINFEI substation.								
	Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes/ No)	Yes, the frequency is in-line to the registered monitoring plan/01/								
	Monitoring Equipment	Electricity meters of the following details have been installed at JINFEI substation. <table border="1"> <tr> <td colspan="3">Make: EDM1 Model: MK6E Accuracy class: 0.2S</td></tr> <tr> <td>Meter</td><td>Main</td><td>Back-up</td></tr> <tr> <td></td><td># 16538506</td><td># 16538507</td></tr> </table> The meter details were cross verified at the time of On-Site Audit and the geotagged meter pictures were shared by the PP from the site/29/.	Make: EDM1 Model: MK6E Accuracy class: 0.2S			Meter	Main	Back-up		# 16538506
Make: EDM1 Model: MK6E Accuracy class: 0.2S										
Meter	Main	Back-up								
	# 16538506	# 16538507								

	Is accuracy of the monitoring equipment as stated in the monitoring plan? If the monitoring plan does not specify the accuracy of the monitoring equipment, does the accuracy of the monitoring equipment comply with local/national standards, or as per the manufacturer's specification?	As per the MR, the accuracy of the EDM1 meter is 0.2s and has also been verified during the site inspection. The calibration frequency of the meters as per the PPA is confirmed to be once in 4 years. The initial calibration of the installed meters was carried out on 22/06/2020, therefore the meters are found to calibrated until 21/06/2024 and no calibration was required during the current monitoring period/09/.
	Is the accuracy valid for the entire measuring range or do different accuracy levels apply to different measuring ranges?	As per the monitoring plan/01/, the accuracy of 0.2S is valid for the entire measuring range. As per PPA para 12.1.4, CEB shall inspect each CEB Meter upon installation and at least once every year thereafter, The periodic test was not conducted during the current MP i.e., 2022 since the same is at the discretion of the CEB these test, however the latest accuracy test was conducted on 14/02/2023 and both the meters were found to be functioning well withing the defined range/10/, therefore the VVB concluded that meters reading obtained during the monitoring period were accurate and thus found acceptable by the VVB.
	Calibration frequency/interval:	The state electricity regulator, CEB is responsible for the certification of CEB Meters through an accuracy test at least once every 4 (four) years thereafter or at any time the readings of Net Energy from the CEB Meter and Seller Back-up Meter differ by an amount greater than 0.5%. The initial calibration conducted on 22/06/2020 is still valid during the monitoring period (valid up to 21/06/2024)/09/.
	Is the calibration interval in line with the monitoring plan and/or	As per the monitoring plan in the registered PD/01/, meter is supposed to be calibrated once in 4 years.

	methodology? If the monitoring plan does not specify the frequency of calibration, is the selected frequency in accordance with the local/national standards, pending until the findings are closed or as per the manufacturer's specifications?	The initial calibration conducted on 22/06/2020 is still valid during the monitoring period (valid up to 21/06/2024)/09/.
	Is the calibration of measuring equipment carried out by an accredited person or institution?	The calibrations are performed by CEB (state regulator).
	Is(are) calibration(s) valid for the whole reporting period?	Yes, it is valid for entire Monitoring Period. The initial calibration conducted on 22/06/2020 is still valid during the monitoring period (valid up to 21/06/2024)/09/. Therefore, it is confirmed that no calibration was required during the current monitoring period.
	Is the calibration carried out for a measuring range comparable with the range for which measurements have been carried out?	Yes, the calibration has been carried out for a measuring range comparable with the range for which measurements have been carried out.
	How were the values in the monitoring report verified?	Cumulative value of $EG_{Facility,y}$ for entire monitoring period is reported in the monitoring report, however the monthly values are reported in the ER calculation sheet/04/. The monthly values were verified from monthly generation and import invoices/28/ issued by Voltas Yellow Ltd and CEB.

		Value of this parameter for the current monitoring period was verified as net 23,764 MWh/28/.
	Does the data management ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	Yes, the adequate QA/QC procedures were followed by the PP. The data transfer process for the said parameter is in accordance with the registered monitoring plan/01/and applied methodology/06/.
Finding	CL 01 and CAR 02 were raised and resolved.	
Conclusion	The parameter has been monitored appropriately, in accordance with the registered monitoring plan (as per measurement methods and procedures to be applied) and applied methodology. The monitoring results were recorded consistently as per the approved frequency in the monitoring plan. The verification team is able to confirm that the project is implemented as per the registered PD and there is no discrepancy observed between the actual monitoring system and the monitoring plan set out in the project description and the applied methodology outlined in the registered VCS PD/01/. The emission reduction calculation for the project activity is based on the electricity supplied by the solar panels. Since 100% data was verified, the assessment team can ascertain that the values taken for emission reduction calculation are free from material errors.	

The table given above describes how each parameter, which is to be measured according to the monitoring plan, has been verified to confirm that the actual monitoring complies with the monitoring plan, monitoring data has been thoroughly assessed and that the sampling requirement are met.

4.6 Non-Permanence Risk Analysis

Not applicable for the project activity.

5 VERIFICATION OPINION

Earthood Services Private Limited (ESPL), contracted by Voltas Yellow Ltd, has performed the independent verification of the emission reductions for the VCS project activity (VCS ID- 2308) grid connected solar PV Project in Mauritius for the monitoring period 01/01/2022 to 31/12/2022 as reported in the Monitoring Report Version 5.0 dated 28/07/2023. Voltas Yellow Ltd is responsible for the collection of data in accordance with the monitoring plan and the reporting of GHG emissions reductions from the project activity.

ESPL commenced the verification based on the baseline and monitoring methodology ACM0002, Version 20.0/06/, the monitoring plan contained in the revised VCS PD Version 1.2 dated 08/09/2020 and VCS guidelines version 4.3. The verification approach of the assessment team is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these.

ESPL planned and performed the verification by obtaining evidence and other information and explanations that ESPL considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

In our opinion the GHG emissions reductions reported for the project activity for the period 01/01/2022 to 31/12/2022 are fairly calculated and stated in the Monitoring Report/03/ Version 3.0 dated 27/06/2023. The GHG emission reductions were calculated correctly on the basis of the approved baseline and monitoring methodology ACM0002, Version 20.0, and the VCS standard/05,06/.

Verification period: From 01/01/2022 to 31/12/2022 (including both days)

Verified GHG emission reductions and removals in the above verification period, broken down by calendar year:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2022	23,557	0	0	23,557
Total	23,557	0	0	23,557

According to ex-ante values, a total of 23,557 tCO_{2e} Emission reductions are estimated for the current MP, and for this Monitoring period the project performed a 18.7% less than expected, it was due to natural meteorological conditions and inverter damage that led to the plant's underperformance which was found acceptable to the assessment team.

Year	Ex-ante emissions reductions /removals	Achieved emissions reductions /removals	Percent difference	Justification for the difference
2022	28,978	23,557	18.7%	The difference can be explained by overall solar PV lower performance due to the: 1) power phase damage of inverters which was identified as a manufacturing defect. 2) Temperature losses, dust and cable transmission losses.

Approved by

Dr. Kaviraj Singh



Managing Director

Date: 03/08/2023

Earthood Services Private Limited

Place: Gurgaon, Haryana

APPENDIX I: ABBREVIATIONS

Abbreviations	Full texts
BE	Baseline Emission
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CEB	Central Electricity Board
CL	Clarification Request
CO ₂	Carbon di oxide
CP	Crediting Period
DR	Desk Review

EB	Executive Board
EF	Emission Factor
ER	External Resource
EIA	Environmental Impact Assessment
ESPL	Earthood Services Private Limited
FAR	Forward Action Request
GHG	Green House Gas
GP	Grouped Project Activity
MR	Monitoring Report
PD	Project Description
PDD	Project Design Document
PP	Project Proponent
PPA	Power Purchase Agreement
tCO ₂ e	Tonnes of Carbon di oxide equivalent
TA	Technical Area
TR	Technical Reviewer
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Units
VVB	Validation and Verification Body

APPENDIX II: REFERENCES

S.N o.	Title of document	Version	Provided by
1.	Registered VCS PD	Version 1.2 dated:08/09/2020	VCS
2.	Registered CDM PDD	Version 1.4, dated:02/05/2019	CDM
3.	VCS Monitoring Report (Final)	Version 5.0 dated 28/07/2023	PP
4.	ER spread sheet	-	PP
5.	a) VCS Standard b) VCS programme guide	Version 4.4 Version 4.3	VCS

6.	Methodology: ACM0002. "Largescale Consolidated Methodology: Grid-connected electricity generation from renewable sources"	Version 20.0	CDM
7.	VCS Project Webpage: https://registry.terra.org/app/projectDetail/VCS/2308	-	VCS
8.	No double counting declaration by PP	Dated: 29/05/2023	PP
9.	Electricity meter calibration report	22/06/2020	CEB
10.	Accuracy Test: Electricity meters annual inspection record conducted by CEB for main meter and check meter:	14/02/2023	CEB
11.	Tool 07: Tool to calculate the emission factor for an electricity system	Version 6.0	CDM
12.	Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation	Version 3.0	CDM
13.	Previous VCS Verification Report (MP3)	Version 3.0 04/11/2022	VCS
14.	Joint Validation and Verification Report	Version 1.0 Dated-20/10/2020	VCS
15.	Geographical co-ordinate for project activity	17/05/2023	PP
16.	Tool on "additionality of first-of-its-kind project activities"	Version 3.0	CDM
17.	Tool for the demonstration and assessment of additionality	Version 7.0	CDM
18.	Manufacturer specification of Solar Panel (China Sunergy Co.Ltd)	2015	PP
19.	EIA License	05/09/2017	PP
20.	Bi-Annual Environmental Monitoring report (FY2022) by GreenYellow Indian Ocean"	30/05/2023	GreenYellow SAS
21.	Bi-Annual Environmental Monitoring report (FY2022) by GreenYellow Indian Ocean"	16/12/2022	GreenYellow SAS
22.	CDM webpage PA 10483 (https://cdm.unfccc.int/Projects/DB/RWTUV1557984446.68/view).	-	PP
23.	Environmental clearance certificate- Solitude 16 MW Solar PV project	07/06/2019	PP
24.	Commercial Operation Certificate	18/12/2019	PP
25.	Line diagram for metering location		PP
26.	Geo tagged plant site photos	-	PP
27.	Employment records	Current MP	PP

28.	Electricity sale and purchase invoices to Central Electricity Board, Mauritius.	2022	PP
29.	Geo tagged photographs: Electricity meters	-	PP
30.	Project Organogram	-	PP
31.	VVB Site Visit records	17/05/2023	VVB
32.	Grievance Logbook	2022	PP

APPENDIX III: COMPETENCE STATEMENT

Competence Statement			
Name	Shifali Guleria		
Education	M.Sc. (Environmental Studies and Resource Management), TERI University		
Experience	3+ year		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	YES (AMS-I.A., AMS-II.G., AMS-II.E., AMS-III.A.V., AMS-I.D, ACM0002)		
Local expert	YES		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (1.2, 3.1)		
Reviewed by	Deepika Mahala	Date	18/02/2022
Approved by	Ashok Gautam	Date	18/02/2022

Name	Shreya Garg
Country	India
Education	M.Sc. (Climate Science & Policy), TERI University
Experience	9 Years +
Field	Climate Change
Approved Roles	
Team Leader	YES
Validator	YES
Verifier	YES

Methodology Expert	AMS.I.A., AMS.I.C., AMS.I.D., AMS.I.F., AMS.II.D., AMS.II.G., AMS.II.J., AMS.III.AV., AMS.III.BL, ACM0002, ACM0012		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.1, TA 1.2, TA 3.1, TA 13.1)		
Reviewed by	Shifali Guleria	Date	21/12/2022
Approved by	Deepika Mahala	Date	21/12/2022

Competence Statement			
Name	Anjali Chaudhary		
Education	Bachelor of technology in Civil Engineering		
Experience	8 months		
Field	Civil Engineering		
Approved Roles			
Team Leader	Yes (VM)		
Validator	Yes		
Verifier	Yes		
Methodology Expert	No		
Local expert	Yes (India)		
Financial Expert	No		
Technical Reviewer	No		
TA Expert (X.X)	No		
Reviewed by	Shifali Guleria (Quality Manager)	Date	11/01/2023
Approved by	Deepika Mahala (Technical Manager)	Date	11/01/2023

Competence Statement	
Name	Kerwyn Donald Dsouza
Education	B. E- Environmental Engineering M. Tech- Renewable Energy Engineering and Management
Experience	7 Years
Field	Renewable Energy, Climate Change, Environment
Approved Roles	
Team Leader	NO
Validator	NO
Verifier	NO

Methodology Expert	NO		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	YES (TA 1.2)		
Trainee	YES		
Reviewed by	Shifali Guleria (Quality Manager)	Date	12/07/2023
Approved by	Deepika Mahala (Technical Manager)	Date	12/07/2023

APPENDIX IV: OVERVIEW FINDINGS

Table 1. Remaining FAR from previous verification

FAR ID	01	Section no.	NA	Date : 16/05/2023
Description of FAR				
PP shall conduct awareness program on the renewable energy and carbon credit to their maintenance team and this shall be checked at the time of next periodic verification.				
Project participant response				Date : 16/06/2023
An awareness program is in place internally for the operation team and all GreenYellow employees since their onboarding (cf. pdf above)				
Documentation provided by project participant				
"Declaration form about the awareness program sAV"				
DOE assessment				Date: 27/06/2023

The assessment team conducted interviews during the onsite audit and questioned the representatives of GreenYellow Indian Ocean regarding the FAR and how the requirements were met by the PP. The assessment team was informed that during the onboarding all the employees were provided an induction training which covered in brief the details on carbon credit mechanism and its applicability on the project activity, the same was confirmed by the VVB during the interviews with the plant personnel and it was concluded that they had the functional understanding of the credit mechanism, positive role of renewable activities and informed contribution towards mitigating GHG emission. The site operations engineered confirmed the details of the training provided as a part of the induction process. The PP has also submitted a declaration that awareness programme is in place for the operations and maintenance team, the document shared by the PP was reviewed by the assessment team and was found to be acceptable.

Thus, the FAR is closed.

Table 2. CL from this verification

CL ID	01	Section no.	4.2	Date : 16/05/2023
Description of CL				
The QA/QC procedure for Monitored parameter: EGfacility,y describes that as per PPA, CEB shall inspect each meter upon installation and at least once every year thereafter. PP shall clarify how the records of these inspections are maintained and mention the same in the MR. PP shall mention the date of last accuracy test conducted for the meters and its validity.				
Project participant response				Date : 16/06/2023
It was agreed by CEB and Voltas Yellow Ltd that the accuracy tests would take place in the second half of 2022, during an operational shut down of the farm. Nonetheless, this didn't happen in 2022 and as a result, the accuracy tests were planned at the earliest, the following year. Hence the latest accuracy tests have been carried out in February 2023.				
Documentation provided by project participant				
"VCS-Monitoring-Report-Solitude_Version 2" "Voltas Yellow - accuracy test on 14 Feb 2023" "Voltas Yellow 2021 report"				
DOE assessment				Date: 27/06/2023
The VVB has reviewed the Accuracy test report shared by the PP, it is confirmed that the accuracies of the two meters are within the standard requirement range. Thus, this finding is closed.				

Table 3. CAR from this verification

CAR ID	01	Section no.	1.1	Date : 16/05/2023
Description of CAR				
1. Monitoring report shall be updated to the latest available version of the template i.e., v4.2 2. The summary description is not in line with the current implementation status of the PA. As per the template guidelines "A summary description of the implementation status of the technologies/ measures (e.g., plant, equipment, process, or management or conservation measure) included in the project." PP shall clarify the reference to proposed status of the project activity in the MR.				
Project participant response				Date : 16/06/2023
1. The monitoring report has been updated to the latest available version of the template. 2. A summary description of the implementation status of the technologies has been included in the monitoring report with information about the panels, inverters, and transformers.				
Documentation provided by project participant				
"VCS-Monitoring-Report-Solitude_Version 2" in track change version				
DOE assessment				Date: 27/06/2023

The PP has shared the latest monitoring report updated to the latest version of the template (v4.2). The revised MR now includes the information about the technologies used in the project activity.
Thus, this finding is closed.

CAR ID	02	Section no.	ER sheet	Date : 27/06/2023
Description of CAR				
Following inconsistencies have been observed from the ER_Spreadsheet SOLITUDE-2022-V1.0- 1. The value of the energy import (kWh) in Tab: Baseline Emissions, cell B6, is found to be inconsistent with the provided invoice for import electricity for the month of April 2022. 2. The value of the energy export (kWh) in Tab: Baseline Emissions, cell D8 is found to be inconsistent with the provided invoice for export electricity for the month of June 2022. PP shall address the inconsistency.				
Project participant response				Date : 27/06/2023
The mentioned inconsistencies are products of human errors and have been corrected, resulting in some changes in the following parameters: EGfacility,y (MWh) & BEy (tCO2) The monitoring report has also been modified accordingly.				
Documentation provided by project participant				
ER_Spreadsheet SOLITUDE-2022-V2.0 – Excel file VCS-Monitoring-Report-Solitude_Version 3				
DOE assessment				Date: 29/06/2023
The PP has made the required corrections and have shared the updated MR and ER sheet. Thus, this finding is closed.				

Table 4.FAR from this verification

FAR ID	NA	Section No.	NA	Date : DD/MM/YYYY
Description of FAR				
NA				
Project participant response				Date : DD/MM/YYYY
NA				
Documentation provided by project participant				
NA				
DOE assessment				Date: DD/MM/YYYY
NA				